

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)
Department of Computer Science and Engineering
CO1 ASSESSMENT TOOL 2 – Agile Sprint Planning & User Story Workshop

Course Code:	CSA10 / CSA1021	Course Title:	Software Engineering
CO Assessed:	CO1	Assessment:	Assessment Tool 2 – Agile Sprint Planning & User Story Workshop
Weightage:	20%	Total Marks:	25
CO1 Statement:	<i>Apply advanced methodologies and agile project management techniques to manage software projects effectively</i>		
Student Name:	Syed Arshad	Reg. No.:	192525060
Date:	10-10-2026	Duration:	60 minutes

Code of Conduct

I Syed Arshad / 192525060 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student:



Annexure A – Agile Sprint Planning & User Story Workshop

Instructions to Students:

Each student is assigned an individual software project problem statement. Based on the given problem statement, perform an **Agile Sprint Planning and User Story Workshop** by applying Agile software engineering practices. Prepare a comprehensive sprint planning document covering the following aspects:

Based on the given problem statement, perform an **Agile Sprint Planning and User Story Workshop** by applying Agile software engineering practices. Prepare a comprehensive sprint planning document covering the following aspects:

- 1. Identify the Project Vision and Stakeholders**
 - Define the project's vision, objectives, and identify all key stakeholders involved.
- 2. Create Epics and User Stories**
 - Break down the requirements into Epics and formulate well-structured User Stories using the format:
As a <user>, I want <feature>, so that <benefit>.
- 3. Define Acceptance Criteria**
 - Write clear and measurable acceptance criteria for each selected user story using the Given–When–Then format or equivalent.
- 4. Prioritize the Product Backlog**
 - Organize and prioritize the product backlog based on business value, customer needs, and project dependencies.
- 5. Plan the Sprint Backlog**
 - Select high-priority user stories for the first sprint and divide them into actionable development tasks.
- 6. Estimate Effort Using Story Points**
 - Assign story points to each selected user story using an appropriate estimation technique (e.g., Fibonacci sequence or Planning Poker) and justify your estimates.
- 7. Present the Sprint Goal and Expected Deliverables**
 - Define a clear sprint goal and describe the expected sprint deliverables, including the features to be completed at the end of the sprint.

Rubrics for Calculation

Criteria	Excellent (5 Marks)	Good (3 Marks)	Satisfactory (2 Marks)	Needs Improvement (1 Mark)
Project Vision & Stakeholder Identification (30%)	Project vision is clear, comprehensive, and aligned with the problem statement. All relevant stakeholders and their roles are correctly identified.	Vision is clear with minor omissions. Most stakeholders are correctly identified.	Vision is partially defined. Some stakeholders are missing or incorrectly identified.	Vision is unclear or incomplete. Stakeholders are poorly identified or missing.
Epics & User Stories (25%)	Epics are well-defined and user stories are complete, follow Agile format, and fully capture functional requirements.	Epics and user stories are mostly complete with minor formatting or requirement issues.	User stories are partially complete with limited Agile structure.	User stories are incomplete, unclear, or do not follow Agile practices.
Acceptance Criteria (20%)	Acceptance criteria are specific, measurable, testable, and aligned with all user stories.	Acceptance criteria are mostly clear and testable with minor gaps.	Acceptance criteria are partially defined and lack clarity or completeness.	Acceptance criteria are missing or not aligned with user stories.
Product Backlog Prioritization & Sprint Planning (15%)	Product backlog is logically prioritized, sprint backlog is realistic, and task breakdown is complete.	Backlog prioritization and sprint planning are mostly appropriate with minor issues.	Prioritization or sprint planning is partially complete with limited justification.	Backlog prioritization and sprint planning are poorly organized or incomplete.
Story Point Estimation, Sprint Goal & Presentation (10%)	Story point estimation is well justified, sprint goal is clear, deliverables are realistic, and presentation is professional and well organized.	Estimation and sprint goal are mostly appropriate with minor inconsistencies. Presentation is clear.	Estimation or sprint goal lacks justification. Presentation is adequate but incomplete.	Estimation is unrealistic or missing. Sprint goal, deliverables, and presentation are unclear or incomplete.

Criteria	Mark
Project Vision & Stakeholder Identification (30%)	/5
Epics & User Stories (25%)	/5
Acceptance Criteria (20%)	/5
Product Backlog Prioritization & Sprint Planning (15%)	/5
Story Point Estimation, Sprint Goal & Presentation (10%)	/5
TOTAL	/25

Real-Time Multi-Facility Corporate ESG Compliance Analytics & Audit Fabric

Problem Statement: Design and develop a highly scalable Corporate ESG Compliance Analytics Platform that tracks resource utilization and carbon footprints across global facility networks. The platform must ingest high-velocity data points—including building smart-meter electricity loads, factory water use metrics, logistics fleet fuel variables, and supply chain waste logs. It must calculate emissions profiles on the fly.

1. Project Vision and Stakeholder Identification

1.1 Project Vision

The vision of the Corporate ESG Compliance Analytics Platform is to provide a centralized, scalable, and reliable software solution for monitoring environmental performance across corporate facilities worldwide. The platform collects environmental data from smart meters, water monitoring systems, logistics fleets, and supply chain waste records. It processes the incoming data and calculates carbon emissions in near real time.

The system helps organizations understand their environmental impact, identify resource wastage, improve sustainability performance, and prepare environmental compliance reports. It also supports data-driven decision-making by presenting environmental information through dashboards, alerts, and analytical reports.

1.2 Project Objectives

1. Collect environmental data from multiple sources across global facilities.
2. Validate and standardize incoming data to improve accuracy.
3. Calculate carbon emissions using appropriate emission factors.
4. Monitor electricity, water, fuel, and waste consumption.
5. Provide real-time dashboards showing resource usage and emissions.
6. Generate reports for management and compliance requirements.
7. Identify abnormal consumption patterns and notify responsible users.
8. Protect environmental data through authentication, access control, and audit logs.
9. Support increasing numbers of facilities and high volumes of incoming data.

1.3 Key Stakeholders

Stakeholder	Role and Responsibilities
Corporate Management	Reviews sustainability performance and makes strategic decisions.
ESG/Sustainability Team	Monitors environmental indicators and prepares ESG reports.
Facility Managers	Monitor electricity and water consumption at individual facilities.
Logistics Managers	Track fleet fuel consumption and related emissions.
Supply Chain Managers	Monitor supplier waste records and environmental performance.
Compliance Officers/Auditors	Verify environmental records and compliance reports.
IT Administrators	Manage infrastructure, security, users, and system availability.

Software Development Team	Designs, develops, tests, and maintains the platform.
Data Engineers	Integrate data sources and maintain data-processing pipelines.

1.4 Stakeholder Expectations

- Management expects clear sustainability summaries and comparisons across facilities.
- ESG teams require accurate emission calculations and report generation.
- Facility and logistics managers need timely information about resource consumption.
- Auditors require traceable records and reliable calculations.
- IT administrators expect a secure, scalable, and maintainable platform.
- Suppliers need a reliable way to submit environmental data.

1.5 Project Scope

The project covers environmental data collection, validation, standardization, carbon-emission calculation, storage, analytics, dashboards, alerts, and compliance reporting. The initial implementation will focus on electricity, water, fleet fuel, and waste data. Integration with every external supplier system, regulatory database, or physical sensor network may be implemented progressively depending on available resources and requirements.

System Architecture:

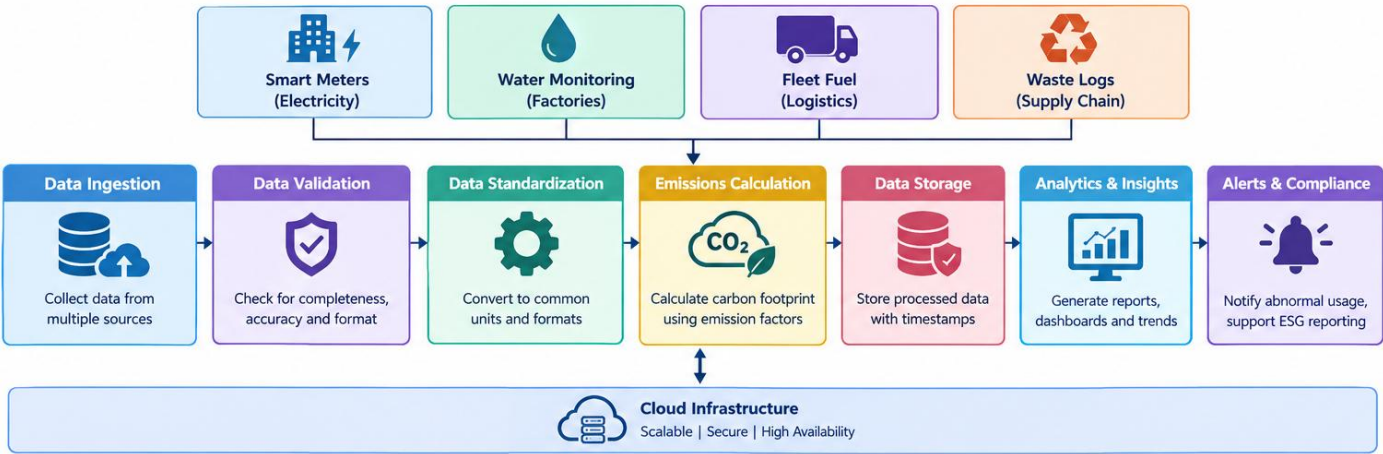


Figure 1: System Architecture of the Corporate ESG Compliance Analytics Platform

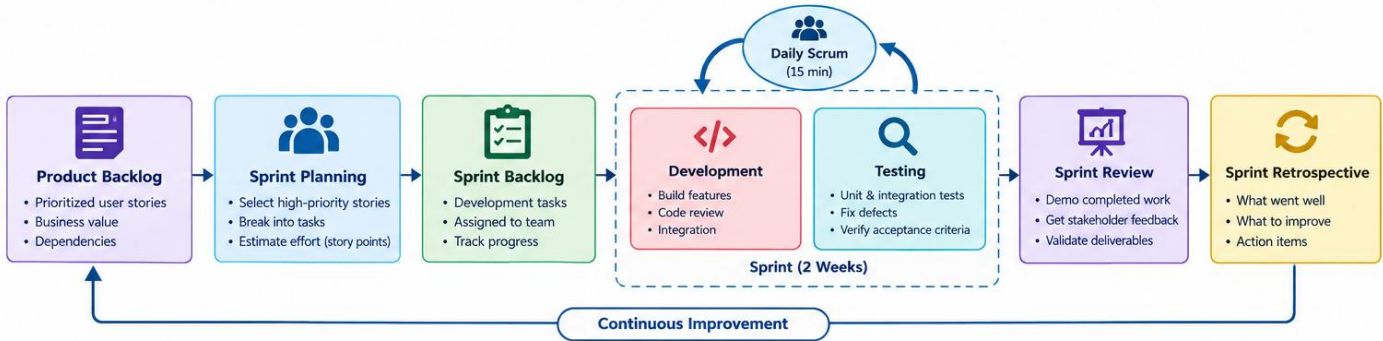


Figure 2: Agile Scrum Sprint Workflow for the ESG Compliance Analytics Platform

2. Create Epics and User Stories

An epic is a large functional requirement that can be divided into smaller user stories. A user story describes a feature from the perspective of a user and explains its benefit.

The user stories below use the required format: As a user, I want a feature, so that I receive a specific benefit.

Epic 1: Environmental Data Ingestion

US-01: Collect environmental data

As a facility manager, I want the platform to collect electricity, water, fuel, and waste data, so that environmental information from different sources can be monitored centrally.

US-02: Validate incoming data

As an ESG analyst, I want the system to validate incoming environmental records, so that incomplete or invalid data does not produce misleading reports.

Epic 2: Data Standardization and Storage

US-03: Standardize environmental units

As a data engineer, I want the system to convert supported measurements into consistent units, so that data from different facilities can be compared accurately.

US-04: Store historical records

As an ESG analyst, I want environmental records to be stored with timestamps and facility identifiers, so that I can analyze historical trends.

Epic 3: Carbon Emissions Calculation

US-05: Calculate emissions automatically

As an ESG analyst, I want the platform to calculate emissions from resource-consumption data using approved emission factors, so that I can monitor carbon footprints without performing manual calculations.

US-06: View emissions by facility

As a corporate manager, I want to compare emissions across facilities, so that I can identify locations with higher environmental impact.

Epic 4: Dashboard and Reporting

US-07: View environmental dashboards

As a sustainability manager, I want a dashboard displaying electricity, water, fuel, waste, and emissions, so that I can understand the organization's environmental performance quickly.

US-08: Generate compliance reports

As a compliance officer, I want to generate reports for a selected period and facility, so that I can review environmental performance and prepare compliance documentation.

Epic 5: Alerts and Data Quality Monitoring

US-09: Receive abnormal-consumption alerts

As a facility manager, I want to receive an alert when resource consumption exceeds a configured threshold, so that I can investigate possible wastage.

US-10: Identify missing data

As a data administrator, I want the system to identify missing or delayed environmental records, so that data gaps can be corrected before reporting.

Epic 6: Security and Auditability

US-11: Manage role-based access

As an IT administrator, I want to assign permissions according to user roles, so that users can access only the functions and information they are authorized to use.

US-12: Maintain audit records

As a compliance auditor, I want important data changes and report-generation activities to be recorded, so that system activities can be reviewed when required.

Epic Summary

Epic	Included User Stories	Business Value
Environmental Data Ingestion	US-01, US-02	Collects reliable environmental data.
Data Standardization and Storage	US-03, US-04	Supports consistency and historical analysis.
Carbon Emissions Calculation	US-05, US-06	Provides emissions information for decisions.
Dashboard and Reporting	US-07, US-08	Improves monitoring and compliance preparation.
Alerts and Data Quality	US-09, US-10	Helps detect wastage and missing records.
Security and Auditability	US-11, US-12	Protects data and supports traceability.

3. Define Acceptance Criteria

Acceptance criteria specify the conditions that must be satisfied before a user story is considered complete. They must be clear, measurable, and testable.

US-01: Collect Environmental Data

- Given a valid environmental record from a supported source, when the platform receives it, then the system must record the measurement, source, timestamp, and facility identifier.
- Given a supported input source is connected, when a valid record arrives, then the record must become available to the processing pipeline.
- The system must identify the data category as electricity, water, fuel, or waste.

US-02: Validate Incoming Data

- Given a record with all required fields and valid values, when validation runs, then the record must be accepted.
- Given a record with a missing required field or invalid value, when validation runs, then the record must be rejected or flagged.
- The system must record a reason for each rejected or flagged record.

US-03: Standardize Environmental Units

- Given a valid measurement in a supported unit, when standardization runs, then the system must convert it to the configured standard unit.
- The original value and unit must remain traceable.
- Unsupported units must be flagged instead of being converted incorrectly.

US-04: Store Historical Records

- Given a valid processed record, when the record is saved, then it must be associated with a timestamp, facility, and resource category.
- When an authorized user selects a facility and date range, then the system must retrieve matching historical records.
- Unauthorized users must not be able to access restricted records.

US-05: Calculate Emissions Automatically

- Given a valid consumption value and an applicable emission factor, when the calculation engine processes the record, then it must calculate emissions using the configured formula.
- The result must identify the input value, emission factor, unit, and calculation period.

- If the required emission factor is unavailable, the system must flag the calculation instead of presenting an unsupported result.

US-06: View Emissions by Facility

- Given emissions data for multiple facilities, when a manager selects a reporting period, then the system must display emissions for each selected facility.
- The displayed totals must match the underlying accepted records and applicable calculation rules.
- The manager must be able to identify the selected period and measurement units.

US-07: View Environmental Dashboards

- Given processed environmental data is available, when an authorized user opens the dashboard, then the system must display the available resource and emissions indicators.
- The dashboard must support filtering by facility and date range.
- The dashboard must indicate when data is missing or has not been updated.

US-08: Generate Compliance Reports

- Given an authorized user selects a facility and reporting period, when the user requests a report, then the system must generate a report using available validated data.
- The report must include the reporting period, facility, resource measurements, emissions totals, and relevant calculation information.
- The system must identify incomplete data where it affects the report.

US-09: Receive Abnormal-Consumption Alerts

- Given a threshold has been configured, when a new valid measurement exceeds it, then the system must create an alert for the responsible user.
- The alert must identify the facility, resource type, measurement, and time.
- Measurements within the configured threshold must not trigger that threshold-based alert.

US-10: Identify Missing Data

- Given an expected record has not arrived within the configured time window, when the data-quality check runs, then the system must flag the missing or delayed record.
- The flag must identify the expected source and facility.
- Once the missing data is received and validated, the status must be updated appropriately.

US-11: Manage Role-Based Access

- Given a user has a defined role, when the user signs in, then the system must allow access only to authorized features and records.
- When an unauthorized user attempts to access a restricted feature, then the system must deny the request.
- Permission changes must be restricted to authorized administrators.

US-12: Maintain Audit Records

- Given an important action such as a data correction or report generation occurs, when the action is completed, then the system must record the relevant user, action, and timestamp.
- Authorized auditors must be able to review available audit records.

4. Prioritize the Product Backlog

The product backlog is an ordered list of features, improvements, and requirements that may be developed for the project. The Product Owner prioritizes items based on business value, user needs, dependencies, risk, and implementation effort.

4.1 Prioritization Method

The MoSCoW method is used to classify requirements:

- Must Have: Essential for the platform's initial working version.
- Should Have: Important, but the system can operate without it temporarily.
- Could Have: Useful enhancement that can be implemented if time permits.

- Won't Have in the First Release: Not included in the current release but may be considered later.

4.2 Prioritized Product Backlog

Priority	ID	User Story / Feature	Category	Reason
1	US-01	Collect environmental data	Must Have	Data collection is the foundation of the system.
2	US-02	Validate incoming data	Must Have	Prevents invalid records from affecting calculations.
3	US-03	Standardize environmental units	Must Have	Makes measurements comparable across sources.
4	US-04	Store historical records	Must Have	Enables retrieval, reporting, and trend analysis.
5	US-05	Calculate emissions automatically	Must Have	Provides the platform's core business function.
6	US-07	View environmental dashboards	Must Have	Allows users to monitor environmental performance.
7	US-06	View emissions by facility	Should Have	Supports comparison between global facilities.
8	US-08	Generate compliance reports	Should Have	Supports environmental reporting and reviews.
9	US-10	Identify missing data	Should Have	Improves data completeness and reliability.
10	US-09	Receive abnormal-consumption alerts	Should Have	Helps identify potential resource wastage.
11	US-11	Manage role-based access	Must Have	Protects sensitive environmental and organizational data.
12	US-12	Maintain audit records	Should Have	Improves traceability and accountability.

Dependency note: Although role-based access is listed near the end of this business-feature ranking, basic authentication and authorization must be included in the initial technical foundation before the platform is deployed to real users. The backlog is a living list and can be reordered as dependencies, risks, and stakeholder feedback become clearer.

4.3 Backlog Prioritization Justification

The initial development effort focuses on collecting, validating, standardizing, and storing data because these activities are prerequisites for reliable emissions calculations. The emissions engine is prioritized next because it provides the platform's central business value. A basic dashboard follows so users can see the processed information.

More advanced reporting, alerts, missing-data monitoring, and audit features can be developed incrementally. However, security and access control must be designed into the platform from the beginning rather than postponed until after deployment.

5. Plan the Sprint Backlog

A sprint backlog is the set of selected product backlog items and the tasks the development team plans to complete during a sprint.

5.1 Sprint Planning Assumptions

- Sprint duration: 2 weeks (10 working days).
- Team size: 4 members for planning purposes.

- Development approach: Agile Scrum.
- Estimation method: Fibonacci story points with Planning Poker.
- Sprint 1 focus: Build the basic environmental data-processing pipeline.
- Sprint 1 capacity assumption: Approximately 20 story points, subject to the team's actual availability and experience.

These are planning assumptions for the assessment, not measured results from an existing implementation.

5.2 Selected Sprint 1 User Stories

The first sprint prioritizes the foundation required for reliable data processing.

User Story Description	Priority	Story Points
US-01 Collect environmental data	Must Have	5
US-02 Validate incoming data	Must Have	3
US-03 Standardize environmental units	Must Have	3
US-04 Store historical records	Must Have	5
US-11 Implement basic role-based access	Must Have	3
Total		19

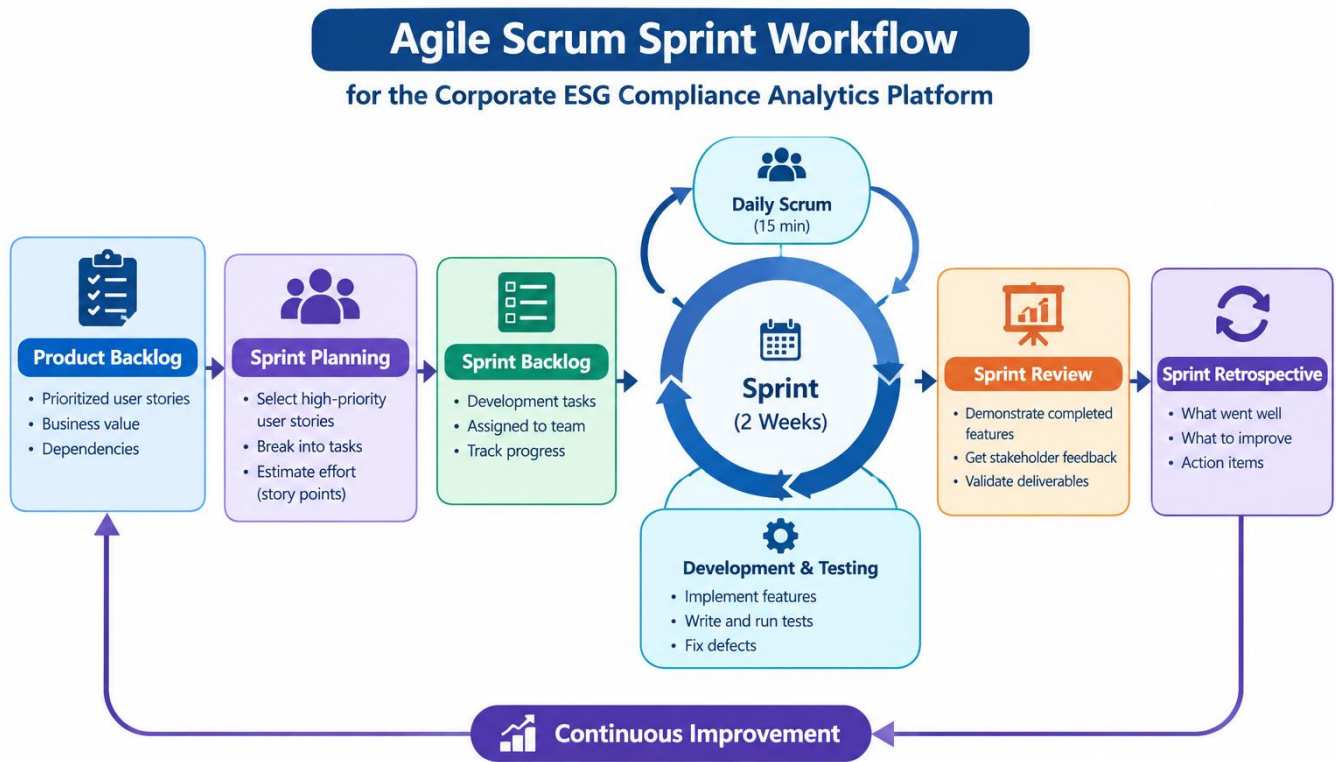
The total of 19 points is within the assumed capacity of 20 points. This leaves a small buffer for integration issues, testing, and defect correction. The team should confirm feasibility after reviewing actual availability and technical complexity.

5.3 Sprint 1 Task Breakdown

User Story	Development Tasks	Expected Output
US-01	Identify required data fields; define input format; implement a data-ingestion endpoint or import mechanism; test sample electricity, water, fuel, and waste records.	Environmental records can enter the processing pipeline.
US-02	Define validation rules; check mandatory fields and data types; handle invalid records; write validation tests.	Invalid or incomplete records are flagged.
US-03	Define standard units; implement supported conversion rules; retain original values; test conversion results.	Supported measurements are converted consistently.
US-04	Design the initial database structure; save validated records; associate timestamps and facility identifiers; test record retrieval.	Processed records can be stored and retrieved.
US-11	Define basic user roles; implement access checks for protected features; test authorized and unauthorized requests.	Basic role-based access is available for protected functions.

5.4 Sprint 1 Schedule

Working Days	Planned Activities
Days 1–2	Sprint planning, data requirements, input formats, database and access design.
Days 3–4	Implement environmental data ingestion and validation.
Days 5–6	Implement unit standardization and historical record storage.
Days 7–8	Implement basic role-based access and integrate the components.
Days 9–10	Execute tests, fix defects, demonstrate completed functionality, and conduct the sprint review and retrospective.



5.5 Definition of Done

A user story is considered complete when:

1. The required functionality has been implemented.
2. The code has been reviewed by another team member.
3. Relevant unit and integration tests pass.
4. Invalid inputs and expected error cases are handled.
5. Access controls are verified for protected functionality.
6. The feature is integrated into the working application.
7. The implementation and known limitations are documented.
8. The feature is demonstrated to the Product Owner or relevant stakeholder.

6. Estimate Effort Using Story Points

Story points are a relative measure of the effort, complexity, uncertainty, and risk involved in completing a user story. They are not direct measurements of working hours.

6.1 Fibonacci Estimation Scale

The Fibonacci sequence is used for estimation because uncertainty tends to increase as tasks become more complex.

Story Points	Meaning	Example
1	Very small	Displaying a simple status field.
2	Small	Adding a basic validation rule.
3	Moderate	Validating a record with several required fields.
5	Medium to complex	Implementing a data-ingestion component.
8	Complex	Building an initial emissions calculation engine with multiple data categories.
13	Very complex or uncertain	Integrating several unfamiliar external systems.

A 13-point story may be broken into smaller stories to make planning and testing easier.

6.2 Story Point Estimates for the Product Backlog

User Story	Complexity Justification	Story Points
US-01: Collect environmental data	Requires input formats, data handling, and source integration.	5
US-02: Validate incoming data	Requires mandatory-field, type, and error checks.	3
US-03: Standardize environmental units	Requires conversion rules and unit tests.	3
US-04: Store historical records	Requires database design, saving, and retrieval.	5
US-05: Calculate emissions automatically	Requires calculation rules, emission factors, units, and validation.	8
US-06: View emissions by facility	Requires aggregation and facility-level filtering.	5
US-07: View environmental dashboards	Requires data retrieval, visual indicators, and filters.	5
US-08: Generate compliance reports	Requires report structure, data aggregation, and output validation.	5
US-09: Receive abnormal-consumption alerts	Requires threshold configuration and alert handling.	5
US-10: Identify missing data	Requires expected-data schedules and gap detection.	3
US-11: Manage role-based access	Requires role definitions and permission checks.	3

US-12: Maintain audit records	Requires event logging and controlled access to records.	3
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6.3 Justification of Estimates

The estimates are illustrative and should be refined by the development team using Planning Poker. In this technique, team members independently select a story-point estimate, reveal their estimates together, and discuss significant differences before agreeing on a final value.

For example, US-01 receives 5 points because data ingestion involves defining input formats, handling multiple resource categories, and testing incoming records. US-02 receives 3 points because validation has a narrower scope. US-05 receives 8 points because the emissions engine must apply the correct emission factor, preserve measurement units, handle missing factors, and produce traceable results.

The estimates may change when the team learns more about external data formats, database requirements, calculation standards, or integration complexity.

7.Sprint Goal and Expected Deliverables

7.1 Sprint 1 Goal

Sprint Goal: To develop a secure and reliable foundation for the Corporate ESG Compliance Analytics Platform by implementing environmental data ingestion, validation, unit standardization, historical record storage, and basic role-based access control.

At the end of the first sprint, the team should be able to demonstrate how sample environmental records enter the system, are validated and standardized, and are stored for future emissions calculations and reporting.

7.2 Expected Sprint 1 Deliverables

1. Data-ingestion component: Accepts sample electricity, water, fleet fuel, and waste records through the agreed input mechanism.
2. Data-validation component: Identifies missing mandatory fields, invalid values, and unsupported input formats.
3. Unit-standardization component: Converts supported measurements into configured standard units while retaining the original values.
4. Database component: Stores validated records with timestamps, facility identifiers, and resource categories.
5. Access-control component: Applies basic role-based permissions to protected functions.
6. Test results: Documents successful test cases, failed test cases, and corrections made.
7. Working prototype: Demonstrates the integrated Sprint 1 functionality using sample data.
8. Sprint review and retrospective: Collects stakeholder feedback and records improvements for the next sprint.

7.3 Sprint Review Checklist

The team can use the following checklist to evaluate the sprint demonstration.

Sprint review verification

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A valid environmental record is accepted.

Invalid or incomplete records are flagged.

Supported measurement units are standardized correctly.

Validated records can be stored and retrieved.

Protected functions enforce the configured user roles.

Relevant tests pass and known defects are documented.

The working prototype is demonstrated to the Product Owner.

Use this checklist during the demonstration. These are planned verification criteria, not claims that the features have already been implemented.

7.4 Expected Future Sprints

The following roadmap shows how the project can continue after Sprint 1.

Sprint	Main Goal	Expected Deliverables
Sprint 1	Establish the data-processing foundation.	Ingestion, validation, unit standardization, storage, basic access control.
Sprint 2	Implement environmental emissions calculations.	Emissions engine, emission-factor configuration, calculation tests, facility-level emissions data.
Sprint 3	Develop monitoring and reporting features.	Environmental dashboard, facility comparisons, reporting-period filters, initial reports.
Sprint 4	Add data-quality monitoring and alerts.	Missing-data detection, consumption thresholds, alerts, audit logging improvements.
Sprint 5	Improve scalability, security, and reliability.	Performance testing, security testing, error handling, integration testing, deployment preparation.
Sprint 6	Complete validation and prepare the release.	User acceptance testing, documentation, defect resolution, stakeholder review, release candidate.

The sprint roadmap is tentative. Sprint planning should be revised according to completed work, stakeholder feedback, technical dependencies, and the team's demonstrated capacity.

8. Agile Roles and Responsibilities

The Scrum framework defines responsibilities that help the team manage the product backlog, development work, and delivery process.

Role	Responsibilities in This Project
Product Owner	Defines the product vision, prioritizes the backlog, clarifies requirements, and reviews completed features.
Scrum Master	Facilitates Scrum events, identifies obstacles, and helps the team follow Agile practices.
Development Team	Designs, develops, integrates, tests, and documents the platform.
ESG Stakeholder Representative	Explains environmental reporting needs and validates emissions-related requirements.
Testing Representative	Reviews acceptance criteria, prepares test cases, and verifies feature Behavior.

In a small academic project, one person may perform more than one responsibility. However, product decisions, development tasks, and testing activities should remain clearly defined.

9. Agile Events and Project Tracking

The team will use the following Scrum activities to monitor progress and improve development.

- **Sprint Planning:** Selects user stories according to priority and team capacity.
- **Daily Scrum:** A short daily meeting to review progress, identify obstacles, and coordinate tasks.
- **Sprint Review:** Demonstrates completed features and obtains stakeholder feedback.
- **Sprint Retrospective:** Identifies what worked well and what should improve in the next sprint.
- **Backlog Refinement:** Clarifies requirements, updates priorities, and breaks large stories into smaller items.

A task board with the columns To Do, In Progress, Testing, and Done can be used to track the work. A story should not be moved to Done until it meets the agreed Definition of Done.

10. Risks and Mitigation During Sprint Planning

Risk	Possible Impact	Mitigation
Inconsistent input formats	Incorrect data processing	Define supported input formats and validation rules.
Incorrect emission factors	Inaccurate carbon calculations	Use approved, documented emission factors and test calculation rules.
Missing or delayed sensor data	Incomplete reports	Flag data gaps and maintain data-quality checks.
Integration difficulties	Sprint delays	Begin with sample records and integrate external systems incrementally.
Excessive sprint workload	Incomplete user stories	Estimate collaboratively and commit only to realistic sprint capacity.
Unauthorized data access	Security and privacy issues	Apply authentication, role-based permissions, and audit logging.
Changing requirements	Rework and delays	Review the product backlog regularly and obtain stakeholder feedback.
Performance problems at high data volumes	Slow processing and dashboards	Conduct load tests and improve the ingestion and storage architecture.

11. Expected Project Outcomes

After the planned sprints are completed, the platform is expected to provide the following benefits:

1. Centralized monitoring of electricity, water, fuel, and waste data.
2. Improved data consistency through validation and unit standardization.
3. Automated carbon-emission calculations using documented emission factors.
4. Facility-level environmental comparisons and historical trend analysis.
5. Dashboards that help management understand environmental performance.
6. Reports that support compliance reviews and internal audits.
7. Alerts that help identify missing data and abnormal resource consumption.
8. Better traceability, access control, and accountability.
9. A modular architecture that can be extended to more facilities and data sources.

These outcomes are intended benefits of the proposed project. Actual performance and accuracy must be confirmed through implementation and testing.

Conclusion

The Agile Sprint Planning and User Story Workshop provides a structured approach to developing the Corporate ESG Compliance Analytics Platform. The project vision establishes the need for a centralized solution to monitor environmental data and calculate carbon emissions across corporate facilities. The identified epics and user stories translate stakeholder requirements into manageable development activities, while acceptance criteria define how each feature will be verified. Product backlog prioritization ensures that essential functions, including data ingestion, validation, standardization, storage, emissions calculation, and access control, receive appropriate attention. Fibonacci story-point estimation supports collaborative effort planning, and the sprint backlog converts selected user stories into actionable development tasks. The first sprint focuses on establishing a reliable data-processing foundation, with subsequent sprints planned for emissions calculations, dashboards, reporting, alerts, and system validation. Regular sprint reviews, testing, stakeholder feedback, and retrospectives will help the team identify issues and improve the product incrementally. Overall, applying Agile Scrum to this project supports organized teamwork, controlled development, continuous improvement, and the gradual delivery of a scalable environmental monitoring and compliance platform.

Furthermore, Agile practices enable the development team to respond effectively to changing environmental regulations, new stakeholder expectations, and emerging technological requirements. By maintaining a prioritized product backlog, the team can focus on the most valuable features and use available time and resources efficiently. Continuous testing and regular sprint reviews help identify defects early and ensure that the developed features satisfy the defined acceptance criteria. The use of collaborative estimation and clear task assignments also improves coordination among team members and reduces the risk of unrealistic commitments. In future releases, the platform can incorporate predictive analytics, automated anomaly detection, and integration with additional environmental monitoring systems to improve sustainability planning. Regular maintenance, security updates, and performance evaluation will help maintain the reliability of the platform as the organization expands. Thus, the combination of Agile project management and software engineering principles can support the successful development of a scalable, reliable, and adaptable ESG compliance solution.